SLIIT COMPUTING | BUSINESS | ENGINEERING	DEPARTMENT OF INFORMATION SYSTEMS ENGINEERING
	FACULTY OF COMPUTING

MODULE OUTLINE			
Module Name	**Information Assurance and Security**		
Module Code	IT3070	Version No.	2017 - 0
Year/Level	3	Semester	2
Credit Points	4		
Pre-requisites	IT1100		
Co-requisites	None		
Methods of Delivery	Lectures (Face-to-face) Tutorials Labs	2 Hours/Week 1 Hours/Week 2 Hours/Week	
Course Web Site	http://courseweb.sliit.lk/		
Date of Original Approval			
Date of Next Review			

MODULE DESCRIPTION		
Introduction	This course will cover the necessary basics of information security and assurance. Course takes a theoretical approach into teaching introductory topics in information assurance and security. This course will expand upon students' knowledge by addressing the essential elements of the domains that comprise a Common Body of Knowledge (CBK) developed by International Information System Security Certification Consortium (ISC)² for information systems security professionals. The course offers an in depth approach to the security process, while providing a framework to prepare for CISSP certification.	
Learning Outcomes	At the end of the module student will be able to:	
	LO1:	Demonstrate the knowledge of basic concepts in information security, including security policies, security models, and security mechanisms.

	LO2:	Understand the structures, transmission methods, transport formats, and security measures used to provide confidentiality, integrity, and availability for information.
	LO3:	Plan for technology development, including risk assessment and evaluate the system design against mission requirements.
	LO4:	Recognize the importance of investigations, incident management, business continuity & disaster recovery.
	LO5:	Recognize legal and ethical aspects of information security.

Assessment Criteria	During the semester, there will be a mid-term examination, one assignment and a final exam. The mid-term test will be based on lecture materials covered until the week before it is held. The final examination will be a comprehensive exam based on the lecture materials covered during the semester.			
	Continuous Assessments			
	• Mid Term Examination	20	%	L01-LO3
	• Assignment	30	%	LO2-LO4
	End Semester Assessment			
	• Final Examination	50	%	LO1-LO5
	TOTAL	100	%	
Estimated Student Workload	Contact Hours			
	• Lecture	26 hours		
	• Tutorial	13 hours		
	• Laboratory	26 hours		
	Time Allocated for Assessments			
	• Continuous Assessments	01 hours		
	• Final Examination	03 hours		
	Reading and Independent Study	131 hours		
	TOTAL	200 hours		
Module Requirement	To pass this module, students need to obtain a pass mark in both “Continuous Assessments” and “End of the Semester Examination” components which would result in an overall mark that would qualify for a “C” grade or above			
Primary References	1. J. M. Stewart, M. Chapple, and D. Gibson, CISSP Certified Information Systems Security Professional Study Guide, 7th. John Wiley & Sons, 2015.			
	2. W. Stallings and L. Brown, Computer Security, Principles and Practice, Pearson, 2014.			

CONTENTS OF THE MODULE

1. **Security and Risk Management**
   - Confidentiality, integrity, and availability concepts
   - Security governance principles
   - Compliance
   - Introduction to information security risk management
   - Security policies, standards, procedures and guidelines

2. **Asset Security**
   - Information and asset classification
   - Ownership (data owners, system owners)
   - Protect privacy
   - Appropriate retention
   - Data security controls
   - Handling requirements (markings, labels, storage)

3. **Security Engineering**
   - Engineering processes using secure design principles
   - Security models fundamental concepts and evaluation models
   - Security capabilities of information systems
   - Security architectures, designs, and solution elements vulnerabilities
   - Web-based and mobile systems vulnerabilities
   - Embedded devices and cyber-physical systems vulnerabilities
   - Cryptography
   - Site and facility design secure principles and physical security

4. **Communication and Network Security**
   - Secure network architecture design (e.g. IP & non-IP protocols, segmentation)
   - Secure network components
   - Secure communication channels
   - Network attacks

5. **Identity and Access Management**
   - Physical and logical assets control
   - Identification and authentication of people and devices
   - Identity as a service (cloud identity)and third-party identity services
   - Access control attacks
   - Identity and access provisioning lifecycle

6. **Security Assessment and Testing**
   - Assessment and test strategies

- Security process data (e.g. management and operational controls)
- Security control testing
- Test outputs (e.g. automated, manual)
- Security architectures vulnerabilities

7. **Security Operations**
   - Investigations support and requirements
   - Logging and monitoring activities
   - Provisioning of resources and foundational security operations concepts
   - Resource protection techniques
   - Incident management and preventative measures
   - Patch and vulnerability management
   - Change management processes
   - Recovery strategies, disaster recovery processes and plans
   - Business continuity planning
   - Personnel safety concerns

8. **Legal and Ethical aspects of Information Assurance**
   - Differentiate between laws and ethics
   - Major national and international laws that affect the practice of information security
   - Intellectual property and DMCA
   - Professional ethics for information security

GENERIC INFORMATION

Any type of plagiarism is not allowed.

Plagiarism: Academic honesty is crucial to a student's credibility and self-esteem, and ultimately reflects the values and morals of the Institute as whole. A student may work together with one or a group of students discussing assignment content, identifying relevant references, and debating issues relevant to the subject. Plagiarism occurs when the work of another person, or persons, is used and presented as one's own.

----------End of Module Outline----------